an amino acid derivative may function through second messenger system like peptide hormones or they may actually enter the cell and function like steroid hormones.

Avoid use of steroid components

The abuse of anabolic steroids can cause serious health problems like high BP, heart diseases, liver damage, cancer, stroke and blood clots. Other side effects of steroid use includes nausea, vomiting, ligament and tendon injuries, head ache, joint pain, muscle cramps, diarrhoea, sleep problem etc.

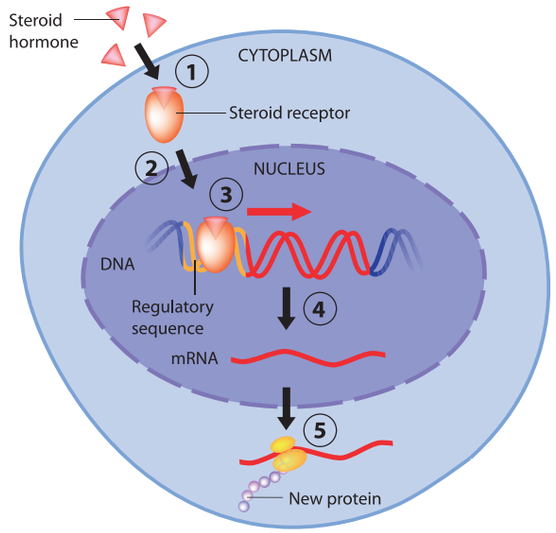


Figure 11.18 Mechanism of steroid hormone action

Summary

Endocrine glands: secrete hormones which diffuse into blood and induce the target organs. They are chemical messengers or organic catalysts which interact with receptor in the target organs.

Hormones speed up or slow down or alter the activities of target organs. The hypo or hyper secretion of hormones leads to serious effects on human beings. Hormones coordinate different physical and mental activities to maintain homeostasis.

Hypothalamus interlinks nervous system and endocrine system. It is located in the diencephalon of cerebrum and controls the pituitary secretion. Pituitary gland secretes six tropic hormones which regulates various physiological functions of our body. Posterior pituitary gland secretes vasopressin that regulates water and electrolyte balance. Oxytocin helps during child birth. Melatonin secreted by pineal gland regulates circadian rhythm of our body. The thyroid gland secretes thyroxine which stimulates the nervous system, skeletal growth, and regulates basal metabolic rate.

Parathyroid gland regulates calcium level in our body. Thymus gland plays a vital role in cell mediated immunity by promoting T lymphocytes maturation. Pancreas regulates blood glucose homeostasis through its secretion of insulin and glucagon.

Adrenal cortex secretes mineralocorticoids which regulates mineral metabolism, glucocorticoids regulates carbohydrate metabolism. Adrenal medulla secretes the hormones adrenalin and noradrenalin. In male, reproductive functions are controlled by testosterone secreted by the testis. In female, ovary secretes three hormones oestrogen, progesterone and relaxin that regulates reproductive functions.

Hormonal deficiency causes serious harmful effects in human. It alters physiological and biochemical functions of the body. This leads into various disorders like acromegaly, dwarfism, tetany, diabetes etc.

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